



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded

Department of Information Technology

In-Semester Examination II

Academic Year: 2025-26 (Semester – II)

B Tech Final Year

Subject: Business Intelligence and Analytics (PEC-IT417)

Date: 23/03/2026

Time: 14.30-15.15

Max. Marks: 20

Answer any four of the Followings.

		Mark	CO	BT
Q.1	Describe the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology.	05	CO1	BT1, 2
Q.2	What are neural networks, and how are they used in Business Intelligence?	05	CO1	BT2
Q.3	Explain how Geographic Information Systems (GIS) contribute to business decision support.	05	CO1	BT3
Q.4	What are Decision Support Systems (DSS)? How do they help in business decision-making?	05	CO2	BT2
Q.5	Define data mining. How is it different from OLAP?	05	CO2	BT3

Course Outcomes:

- 1) Demonstrate understanding of core AI concepts and techniques.
- 2) Implement and apply various AI algorithms effectively.
- 3) Develop AI-driven solutions to real-world problems.
- 4) Evaluate ethical and social implications of AI.
- 5) Conduct AI research and contribute to advancements in the field.

Bloom's Taxonomy

BT1-Remember,
BT2-Understand,
BT3- Apply,
BT4-Analyze,
BT5-Evaluate,
BT6-Create

	SGGS Institute of Engineering and Technology, Nanded	
	ISE 2 Examination 2025-26 (Semester-II)	
	Final Year B. Tech. – IT	
	Subject: Elective - Artificial Intelligence	Code: PEC-IT416
Date: 23/03/2026	Time: 14.30 To 15.15	Max. Marks: 20

Notes:

1. SOLVE ANY FOUR QUESTIONS. 2. This paper contains ONE page.
3. Figures to right indicate full marks. 4. Assume suitable data if necessary and mention it.
5. No calculator is permitted.

Q. No.	Question	Marks	CO/BT
Q. 1	Explain Horn and definite clauses.	[05]	CO2/ BT2
Q. 2	Which of the following are correct? a. $\text{False} \models \text{True}$ b. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B)$ c. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B) \wedge (\neg D \vee E)$ d. $(A \vee B) \wedge \neg (A \Rightarrow B)$ is satisfiable e. $A \Leftrightarrow B \models \neg A \vee B$	[05]	CO2/ BT4
Q. 3	Write a short note on Local search algorithm.	[05]	CO2/ BT1
Q. 4	Complete the following exercises about logical sentences: a. Translate into good, natural English (no x s or y s!): $\forall x, y, l$ $\text{SpeaksLanguage}(x, l) \wedge \text{SpeaksLanguage}(y, l)$ $\Rightarrow \text{Understands}(x, y) \wedge \text{Understands}(y, x)$. b. Explain why this sentence is entailed by the sentence $\forall x, y, l \text{ SpeaksLanguage}(x, l) \wedge \text{SpeaksLanguage}(y, l)$ $\Rightarrow \text{Understands}(x, y)$. c. Translate into first-order logic the following sentences: (i) Understanding leads to friendship. (ii) Friendship is transitive. Remember to define all predicates, functions, and constants you use.	[05]	CO2/ BT5,6
Q. 5	Which of the following are valid (necessarily true) sentences? a. $(\exists x x = x) \Rightarrow (\forall y \exists z y = z)$. b. $\forall x P(x) \vee \neg P(x)$. c. $\forall x \text{ Smart}(x) \vee (x = x)$.	[05]	CO2/ BT5

Course Outcomes (CO):

- CO1: Demonstrate understanding of core AI concepts and techniques.
CO2: Implement and apply various AI algorithms effectively.
CO3: Develop AI-driven solutions to real-world problems.
CO4: Evaluate ethical and social implications of AI.
CO5: Conduct AI research and contribute to advancements in the field.

Bloom's Taxonomy (BT):

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate BT6- Create



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**Department of Information Technology
Academic Year: 2025-26 (Semester II)
In-Semester Examination II**

Class: Final Year B.Tech.	Course Name: Affective Computing	Course Code: PEC-IT419
Date: 23.03.2026	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Question	Marks	CO	BT
Q.1	Define emotion elicitation. Explain the difference between Posed, Naturalistic, and Induced emotional expressions with suitable examples.	05	CO2	BT2
Q.2	Describe the role of the Institutional Review Board (IRB) in emotion research experiments. List and briefly explain the five key documents required before data collection begins.	05	CO2	BT3
Q.3	Apply the five-stage facial expression recognition pipeline to a scenario where a classroom engagement monitoring system is being built. Identify the function of each stage (Face Detection → Tracking → Landmark Detection → Feature Extraction → Classification).	05	CO5	BT6
Q.4	Compare Geometric, Appearance (HOG/SIFT), and Spatio-Temporal feature extraction techniques for Automatic Facial Expression Recognition. Analyze the trade-offs in terms of computational complexity and accuracy.	05	CO4	BT5